Problem 1: Array – Find the Missing Number

Difficulty:

Easy

Problem Statement

You are given an array containing **n distinct numbers** taken from the range **0 to n**. Exactly one number is missing.

Write a program to find the missing number.

Example 1

Input

nums = [3,0,1]

Output

2

Example 2

Input

nums = [0,1]

Output

2

Constraints

- $1 \le n \le 100000$
- All elements are unique.

Expected Complexity

- Time: **O(n)**
- Space: **O(1)**

Problem 2: String – First Non-Repeating Character

Difficulty

Medium

Problem Statement

Given a string, find the **first character that appears only once**.

If no such character exists, return -1.

Example 1

Input

s = "leetcode"

Output

l

Example 2

Input

s = "aabbcc"

Output

-1

Example 3

Input

s = "swiss"

Output

w

Constraints

- String contains lowercase English letters.
- Length ≤ 100000.

Expected Complexity

- Time: **O(n)**
- Space: **O(1)**

Problem 3: Array + String – Group Anagrams

An anagram is a word or phrase made by mixing up the letters of a different word or phrase. You must use all the original letters the exact same number of times

Difficulty

Medium-Hard

Problem Statement

Given an array of strings, group all the anagrams together.

Two strings are anagrams if they contain the same characters in a different order.

Example

Input

["eat","tea","tan","ate","nat","bat"]

Output

[

["eat","tea","ate"],

["tan","nat"],

["bat"]

]

The order of groups does not matter.

Constraints

- 1 ≤ number of strings ≤ 10000
- Each string contains only lowercase English letters.

Expected Complexity

- Time: **O(n × k log k)**, where:
  - **n** = number of strings
  - **k** = average length of a string
- Space: **O(n × k)**